Second, suppose that we misspecify the vector S of size weights—for example, by defining $z_t = \Sigma_i S_i^\circ \check{u}_{it}^e$ using a wrong vector S°. Then, the parameters estimated using IV are still consistently estimated but the parameters estimated using OLS can be biased.[20] After all, (9) still holds—namely, $\mathbb{E}[(p_t - \psi y_{St})z_t] = 0$—so that the IV procedure in proposition 1 remains valid.

Third, if we assume that the elasticities are homogeneous across entities (of demand in our model in [1] and [2]) while they are actually heterogeneous, then the IV and OLS estimates are the equal-weighted averages of coefficients when N is large. For instance, the IV estimators yield estimates of ψ and ϕ_E^d, and the OLS estimators estimate coefficients corresponding to the interpretation that the elasticity of demand is ϕ_E^d instead of ϕ_S^d (see app. sec. D.8).

The limits of GIVs, threats to identification, and diagnostics for misspecification.—We now discuss the main limits to the applicability of GIVs, threats to identification, and some diagnostics for misspecification.

Conditional on GIVs being sufficiently powerful, the most important threat to identification is that we do not properly control for common factors. Indeed, calling λ_Γ^e and η_t^e the estimated values, $z_t = u_{\Gamma t} + \check{\lambda}_\Gamma \check{\eta}_t - \check{\lambda}_\Gamma^e \check{\eta}_t^e$, so there is a danger that, even after controlling for $\check{\eta}_t^e$ in the regression, we will not completely eliminate the error term $\check{\lambda}_\Gamma \check{\eta}_t - \check{\lambda}_\Gamma^e \check{\eta}_t^e$.[21] This bias is larger when $|\check{\lambda}_\Gamma|$ is greater—that is, when loadings are correlated with size. Indeed, omitting factors for which $\check{\lambda}_\Gamma = 0$ is inconsequential. We provide four concrete suggestions to mitigate this concern.

First, we can add factors for which we estimate the loadings; see section IV.B. As we show, when T is sufficiently large, we accurately measure $\check{\lambda}$ and $\check{\eta}_t$ and this procedure detects (strong) missing factors. While one cannot rule out weak factors, it is reassuring if the GIV estimates remain stable when adding factors. A related concern is that the factor loadings are unstable over time. In this case, it is possible to use more advanced methods to extract factors—for instance, by modeling $\lambda_{it} = X_{it}\dot{\lambda}$, as we discuss in appendix section D.12. Identification then depends on the correct specification of X_{it}.

Second, we can "narratively check" the idiosyncratic shocks that are being used in the GIV. Indeed, if granular shocks have aggregate consequences, it is often the case that the researcher can "label" them and understand them—especially if we have high-frequency data. This conforms

[20] Calling $\xi = \mathbb{E}[z_t u_{S^\circ t}]/\mathbb{E}[z_t^2]$ (which is one when $S^\circ = S$, assuming [35]), then the OLS above gives the estimates (for large T) $\mu^e = \mu\xi$ and $M^e = M\xi$. For some selection procedures (e.g., selecting the shocks to some prespecified entities as we discussed), we still have that $\xi = 1$, so that OLS is still valid. However, if $S^\circ = S + \epsilon^S$, where ϵ^S is a vector of measurement error, then typically $\xi \neq 1$ and the OLS procedure is biased.

[21] As we do control for $\check{\eta}_t^e$ in the regression, the bias is due to the residual of $\check{\lambda}_\Gamma \check{\eta}_t - \check{\lambda}_\Gamma^e \check{\eta}_t^e$ after controlling for $\check{\eta}_t^e$.

to the spirit of the credibility revolution that the researcher is very clear about the underlying "assignment-to-treatment process"; here we can be clear about what the "treatments" are. The researcher can check the top, say, 10 events and confirm using additional information that those shocks indeed are valid idiosyncratic shocks. It is then possible to construct the GIV based only on those narratively checked shocks.[22] This analysis also helps researchers to provide economic content to the variation used in estimating the parameters of interest, which lends further credibility to the estimates.

Third, the narrative check may reveal that a large idiosyncratic shock coincides with a large aggregate event. For instance, there may be an important policy announcement in the sample. If this happens once in the sample, it is hard to detect using standard factor models. Such "sporadic factor realizations" can lead to bias, and it is therefore prudent to remove these dates.

Fourth, we can do an overidentification test. We can construct two GIVs based on two types of entities—for example, developing and developed countries. We then form the GIVs $z_t^{(1)} = S^{(1)\prime}\check{u}_t$ and $z_t^{(2)} = S^{(2)\prime}\check{u}_t$ based on the size-weighted sum of idiosyncratic shocks of each type (i.e., with two different sets of weights $S^{(1)}$ and $S^{(2)}$) and test whether the estimates using either instrument are the same. If the estimates are significantly different, then this points to misspecification—for instance, of the factor model or that the elasticities are heterogeneous across entities. In this case, we can explore generalization in terms of the factor model (see sec. IV.B) or the model of elasticities (see sec. IV.A).

What is an idiosyncratic shock?—Mathematically, an idiosyncratic shock is plainly a random variable u_{it} such that $\mathbb{E}_{t-1}[(\eta_t', \varepsilon_t)u_{it}] = 0$. That said, it is useful to discuss different types of economic settings that map into this definition. In some cases, it is quite clear—for example, a random productivity or demand shock. But there are more subtle types of idiosyncratic shocks. One is an "unexpected change in the loading on a common shock." For instance, if China decreases oil consumption by more than anticipated in response to a global economic downturn, then it is an idiosyncratic shock. Formally, if demand is $y_{it} = \phi^d p_t + (\lambda_i + \tilde{\lambda}_{it})\eta_t + v_{it}$, with $\mathbb{E}_{t-1}[(1, \eta_t', \varepsilon_t)\tilde{\lambda}_{it}] = 0$, then $u_{it} := \tilde{\lambda}_{it}\eta_t + v_{it}$ is a valid idiosyncratic shock.

[22] This is roughly what the "narrative" approach in the literature does (e.g., Caldara, Cavallo, and Iacoviello 2019). But the GIV procedure helps researchers even in the narrative context, since it automates the "preselection" of the top K (perhaps $K = 10$) shocks, by selecting the events with the largest K values in $S_i|\check{u}_{it}|$. Hence, researchers do not need to know the whole history before selecting their main events—the GIV gives them the most promising candidate events, and the detailed historical search is simply restricted to K events. In addition, the factor analysis in the GIV gives controls η_t^e that are usable when running regressions, which increases the precision of estimators.

The volatility of idiosyncratic shocks can depend on the common shocks. For instance, suppose that $u_{it} = \sigma_t v_{it}$, where σ_t and (η_t', ε_t) could be correlated (e.g., σ_t could increase when $|\varepsilon_t|$ is high), but $\mathbb{E}_{t-1}[(\eta_t', \varepsilon_t)\sigma_t v_{it}] = 0$ (a sufficient condition is that v_{it} is independent of $\sigma_t(\eta_t', \varepsilon_t)$); then, u_{it} is an idiosyncratic shock because $\mathbb{E}_{t-1}[(\eta_t', \varepsilon_t)u_{it}] = 0$.[23]

IV. Extensions

We now present a succession of extensions of the basic GIV procedure that cover a range of empirically relevant cases. The online appendix gives a number of other extensions.

A. *Heterogeneous Demand Elasticities*

We have assumed so far that demand elasticities are constant across entities. We now extend the model to the case where demand elasticities vary across entities and are a function of characteristics x_i.[24] The parameter x_i is a k-dimensional vector, and the first entry is equal to one.[25] We represent the demand elasticities as

$$\phi_i^d = x_i \dot{\phi}^d = \sum_{\ell=1}^{k} x_{i\ell} \dot{\phi}_\ell^d, \tag{47}$$

for some k-dimensional vector $\dot{\phi}^d = (\dot{\phi}_\ell^d)_{\ell=1,\dots,k}$ that is to be estimated. With x the $N \times k$ matrix of characteristics, we summarize the elasticities as $\phi^d = x\dot{\phi}^d$. We assume that λ spans x. For instance, we can have $\lambda = (x, \hat{\lambda})$, where $\hat{\lambda}$ comprises loadings orthogonal to x. The following proposition describes how we can consistently estimate ψ and $\dot{\phi}^d$.

Proposition 6 (Estimation of heterogeneous parametric elasticities). Let assumptions 1 and 3 hold. Consider the model with heterogeneous elasticities of demand following (47). Define $R^x := (x'x)^{-1}x'$, $\dot{y}_t := R^x(y_t - C_t^y m^y)$ (which has dimension k), $\check{u}_t := Q^\lambda(y_t - C_t^y m^y)$, and $z_t := \Gamma'\check{u}_t$, where Γ satisfies (4). We then identify ψ and $\dot{\phi}^d$ using the moment conditions $\mathbb{E}[(p_t - \psi y_{St})z_t] = 0$ and $\mathbb{E}[(\dot{y}_t - \dot{\phi}^d p_t)z_t] = 0_{K\times 1}$—that is,

$$\mathbb{E}\left[\left(\dot{y}_{\ell t} - \dot{\phi}_\ell^d p_t\right)z_t\right] = 0 \qquad \text{for } \ell = 1, \dots, k. \tag{48}$$

This procedure extends the model with homogeneous demand elasticities, where the moment condition (48) simplifies to $\mathbb{E}[(y_{Et} - \phi^d p_t)z_t] = 0$.[26]

[23] A related concern is that some (small) clusters of the u_{it}'s are correlated. In that case, we can either aggregate the entities within each cluster or, alternatively, assume V^u from assumption 4 to be block diagonal.

[24] This x_i might be different from X_i in assumption 1; e.g., it could be a strict subset of it.

[25] Appendix section D.9 considers a nonparametric version, which is more involved.

[26] Indeed, when the elasticity is the same across entities, $x = \iota$ so that $R^x = E'$ and $\dot{y}_t = y_{Et}$.

Computing $\dot{y}_t$ and $\check{u}_t$ in proposition 6 corresponds to running cross-sectional regressions of $y_{it} - C_{it}^y m^y$ on x_i and $\check{\lambda}_i$, for every date t (and given m^y):

$$y_{it} - C_{it}^y m^y = x_i \dot{y}_t + \check{\lambda}_i \check{\eta}_t + \check{u}_{it}, \tag{49}$$

which yields an exact recovery (as in lemma 1) of $\dot{y}_t$, $\check{\eta}_t$, and $\check{u}_{it}$. Then, we form the GIV $z_t := \Sigma_i S_i \check{u}_{it}$ and use the moment conditions in proposition 6 to recover the elasticities. One can add controls, as in proposition 4, and the procedure is otherwise the same.

B. *When the Factor Loadings Are Estimated*

The results so far are derived by imposing assumption 1 that we know λ. We now show how the procedure extends when we relax this assumption and estimate the factor loadings. We remain in the case where $T \to \infty$ and N is fixed.[27]

Recall that we have $\lambda = (\iota, \check{\lambda})$. As in all factor models, we need a normalization. We choose the normalization that $(1/N)\lambda'\lambda = I_r$ and that $V^{\check{\eta}}$ is diagonal.[28] To ensure uniqueness of the representation, we also impose that the diagonal terms of $V^{\check{\eta}}$ are distinct and in decreasing order; this is a purely technical condition without economic substance that could be relaxed at the cost of distracting notations. This leads to the following system of moment conditions.

Proposition 7 (GIV estimation with controls and estimation of factor loadings λ). Let assumption 3 hold. Define $\check{y}_{it} := y_{it} - y_{Et}$ and similarly $\check{C}_{it}^y := C_{it}^y - C_{Et}^y$. Given candidate values m^y and $\lambda = (\iota, \check{\lambda})$, we construct estimates of the factors using $\check{\eta}_t(m^y, \check{\lambda}) := R^{\check{\lambda}}(y_t - C_t^y m^y)$ and the associated GIV $z_t(m^y, \check{\lambda}) := S' Q^{\check{\lambda}}(y_t - C_t^y m^y)$. Define θ to be $(\psi, \phi^d, m^p, b^p, m^y, b^y, \check{\lambda})$ and θ_T^e to be the generalized method of moments (GMM) estimator of θ associated with the following moment conditions:

$$\sum_t \left(-\check{y}_t + \check{\lambda}\check{\eta}_t(m^y, \check{\lambda}) + \check{C}_t^y m^y\right)\check{\eta}_t(m^y, \check{\lambda})' = 0, \tag{50}$$

$$\sum_t \left(-\check{y}_t + \check{C}_t^y m^y\right)' \check{C}_t^y = 0, \tag{51}$$

and moments (29) and (30), making $\check{\eta}_t$ and z_t depend on $(m^y, \check{\lambda})$. Then, assuming that $\psi \neq 0$ and the regularity assumption 5 from appendix A, the vector θ_0 is consistently estimated (with $T \to \infty$ and fixed N), and we have $\sqrt{T}(\theta_T^e - \theta_0) \to N(0, V^\theta)$ for a matrix V^θ.

[27] Banafti and Lee (2022) extend the results in this paper by studying the case where T, $N \to \infty$.

[28] To remove the classic indeterminacy that the fth factor λ^f could be changed into $-\lambda^f$, we can impose, e.g., that for each f, the first nonzero λ_i^f is positive.